

IMAT MOCK EXAM 8 — "Final Rehearsal"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 44/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Logical Reasoning & Reading (Q1-9)

1. "The human gut hosts trillions of microbes that assist digestion, synthesise vitamin K and several B vitamins, and help train the developing immune system. A course of antibiotics can transiently reduce the diversity of this community. In mice, transplanting gut microbes from obese donors transfers some of the donors' metabolic traits to lean recipients, and diet can reshape the community's composition within days."

Which of the following is not supported by the passage?

- A) Gut microbes influence their host's metabolism.
- B) Diet can change the composition of the gut community within days.
- C) Gut microbes synthesise some vitamins.
- D) Taking probiotic supplements cures inflammatory bowel disease.
- E) Antibiotics can temporarily reduce gut microbial diversity.

Answer: D

The passage explicitly states that gut microbes synthesise vitamin K and several B vitamins, that antibiotics transiently reduce diversity, that transplants transfer metabolic traits between mice, and that diet reshapes the community within days. It makes no claim anywhere about treating or curing inflammatory bowel disease with probiotic supplements, so that statement goes beyond the text. The probiotic claim is plausible-sounding health advice, but plausibility is not support: nothing in the passage links probiotics to any disease outcome.

2. A city finds that neighbourhoods with more street trees have lower childhood asthma rates. The council concludes that planting trees will reduce asthma.

Which of the following, if true, most weakens the council's conclusion?

- A) Tree-lined streets are concentrated in wealthier neighbourhoods, which also have better housing and healthcare.
- B) Asthma has many triggers.
- C) Trees produce pollen.
- D) Trees are planted on wide streets.
- E) Some neighbourhoods have parks rather than street trees.

Answer: A

The council infers causation from a correlation. If tree-lined streets are concentrated in wealthier neighbourhoods, then wealth plausibly drives both the tree planting and the lower asthma (better housing, healthcare, lower pollution exposure), so planting more trees need not change asthma rates at all: this confounder attacks the inference directly. By contrast, the fact that trees produce pollen, or that asthma has many triggers, does not explain why tree-rich areas show less asthma, so neither undermines the conclusion as strongly.

3. (General knowledge) The Milano Cortina 2026 Winter Olympic Games are the __ time that Italy has hosted the Winter Olympic Games.

- A) fourth B) first C) second D) fifth E) third

Answer: E

Italy has previously hosted the Winter Games at Cortina d'Ampezzo in 1956 and in Turin in 2006. Milano Cortina 2026 is therefore the third Italian Winter Olympics. Calling it the first or second would ignore one of those earlier editions.

4. A sealed box contains a number of levers. Exactly one of the following two statements is true:

- (1) All of the levers in the box are made of steel.
(2) None of the levers in the box is made of steel.

Which conclusion follows necessarily?

- A) No lever is made of steel.
B) The box contains at least one lever.
C) All the levers are made of steel.
D) Some, but not all, of the levers are made of steel.
E) The box is empty.

Answer: B

Givens: exactly one of the two statements is true. Statement 1: all levers in the box are steel.
Statement 2: no lever in the box is steel.

If the box were empty, both statements would be true at once (a universal claim over zero levers holds vacuously), contradicting exactly one true.

So the box contains at least one lever.

With at least one lever: if all are steel, statement 1 is true and statement 2 false; if none is steel, statement 1 is false and statement 2 true. Both cases satisfy the condition, so which statement is the true one stays undetermined.

Result: the only conclusion forced in every allowed case is that the box holds at least one lever.

The tempting claim that the box is empty is exactly the case ruled out by the first deduction.

5. Three tasks (triage, laboratory and records) must be assigned, one each, to Anil, Bea and Cara. Anil refuses to do triage. Cara refuses to do records.

How many different valid assignments are possible?

- A) 2 B) 3 C) 4 D) 5 E) 6

Answer: B

Givens: three tasks (triage, laboratory, records) assigned one each to Anil, Bea and Cara; Anil refuses triage; Cara refuses records.

Count by Cara's choice, since her refusal touches only records.

Case 1: Cara takes triage. Laboratory and records go to Anil and Bea in either order: 2 assignments.

Case 2: Cara takes laboratory. Triage and records remain; Anil refuses triage, so Anil takes records and Bea takes triage: 1 assignment.

Case 3 is impossible: Cara refuses records.

Result: $2 + 1 = 3$ valid assignments

The tempting 6 counts every permutation and ignores the two refusals; the tempting 2 stops after the first case.

6. A clock reads 7:35. What is the angle between the hour hand and the minute hand?

- A)** 12.5° **B)** 17.5° **C)** 20° **D)** 22.5° **E)** 35°

Answer: B

Given: time 7:35; minute hand speed 6 degrees per minute; hour hand speed 30 degrees per hour plus 0.5 degrees per minute.

Relationship: measure each hand clockwise from 12, then subtract the two angles.

Substitute (minute hand): $35 \times 6 = 210$ degrees

Substitute (hour hand): $7 \times 30 + 35 \times 0.5 = 210 + 17.5 = 227.5$ degrees

Difference: $227.5 - 210 = 17.5$ degrees

Result: 17.5 degrees between the hands.

Classic slip: 35 degrees comes from comparing 210 with $7 \times 30 = 210$ and forgetting that the hour hand has drifted 17.5 degrees past the 7.

7. In a dose-finding trial, patients rated their average pain on a 0 to 10 scale. At doses of 0, 2.5, 5 and 10 mg the mean pain scores were 8, 5, 3 and 3 respectively.

Which conclusion is best supported?

- A)** Higher doses are always better.
B) The drug is unsafe at 10 mg.
C) Increasing the dose beyond 5 mg produces no further average reduction in pain.
D) The drug has no effect on pain.
E) 10 mg is the optimal dose.

Answer: C

Givens: mean pain scores 8, 5, 3, 3 at doses 0, 2.5, 5 and 10 mg.

From 0 to 2.5 mg the mean falls $8 \rightarrow 5$; from 2.5 to 5 mg it falls $5 \rightarrow 3$.

From 5 to 10 mg the mean stays at 3: the dose-response curve has flattened.

Result: the supported conclusion is that increasing the dose beyond 5 mg produces no further average reduction.

The trial recorded only pain scores, so the data contain no safety information and cannot identify an optimal dose.

The claim of no effect fails because the mean dropped from 8 to 3, and higher doses are not always better because the fall stops.

8. A share price rises by 10% one day and falls by 10% the next day. The net change over the two days is:

- A)** a 1% decrease **B)** no change **C)** a 1% increase **D)** a 10% decrease **E)** a 19% decrease

Answer: A

Given: a rise of 10 percent on day 1 and a fall of 10 percent on day 2; take a starting price of 100.

Relationship: successive percentage changes multiply as factors of the running price.

Substitute: $100 \times 1.10 = 110$ after the rise

$110 \times 0.90 = 99$ after the fall

Result: 99 against a start of 100, a net decrease of 1 percent.

Cross-check: $1.10 \times 0.90 = 0.99$, which is 99 percent of the start, confirming the 1 percent loss.

Classic slip: adding 10 and -10 to reach no change forgets that the fall acts on the raised price 110, not on the original 100.

9. In a class of 30 students, 18 study biology, 15 study chemistry, and 8 study both. How many study neither subject?

- A) 2 B) 3 C) 4 D) 5 E) 6

Answer: D

Given: class of 30; biology 18, chemistry 15, both 8.

Relationship: inclusion-exclusion, at least one subject = biology + chemistry – both, because the 8 shared students are counted twice in the raw sum.

Substitute: $18 + 15 = 33$, then $33 - 8 = 25$ students study at least one subject.

Neither = class total – union = $30 - 25 = 5$

Result: 5 students study neither subject.

Classic slip: 4 comes from computing $30 - 18 - 8$, which treats the shared students as if they stood outside both subjects; the overlap belongs inside the union and is subtracted once when the union is formed.

Section B - Biology (Q10-32)

10. A father is blood group AB; a mother is blood group O (genotype OO). Which statement about their children is correct?

- A) All children are AB.
B) The four ABO phenotypes are all equally possible.
C) All children are O.
D) The children's groups cannot be predicted.
E) Each child must be A or B (genotypes AO or BO); AB and O are impossible.

Answer: E

Givens: father AB, so his gametes carry A or B; mother OO, so her gametes carry O only. Each child gets one allele from each parent.

Father's contribution: A with probability one half, B with probability one half. Mother's contribution: O always.

Combine: A + O gives AO (group A); B + O gives BO (group B), each with probability one half.

Result: every child is group A or group B, and groups AB and O are impossible in this pairing.

AB needs A and B together, but the mother can never supply either; O needs O from both parents, but the father never supplies O.

The tempting claim that the children's groups cannot be predicted fails: the outcome set is fully determined, only the A-or-B split is fifty-fifty.

11. Genes A and B lie on the same chromosome and crossing over between them never occurs. An individual with chromosomes AB/ab (coupling arrangement) is crossed with ab/ab. The offspring will be:

- A)** only the two recombinant types
- B)** offspring in a 9:3:3:1 ratio
- C)** only the two parental types, AB/ab and ab/ab, in equal proportions
- D)** all offspring AB/ab
- E)** all four phenotypic classes in a 1:1:1:1 ratio

Answer: C

Given: genes A and B on the same chromosome, no crossing over between them; cross AB/ab × ab/ab.

No crossing over means each homologue is passed on intact: the AB/ab parent makes exactly two gamete types, AB and ab, each with probability one half.

The tester ab/ab makes only ab gametes.

Combine: AB + ab gives AB/ab; ab + ab gives ab/ab, in a 1:1 ratio.

Result: only the two parental types appear, in equal proportions.

The classic 1:1:1:1 testcross ratio needs the recombinant gametes Ab and aB, which cannot be produced without crossing over.

12. A woman is a carrier of haemophilia ($X^H X^h$). Her husband has haemophilia ($X^h Y$). What fraction of their daughters is expected to have haemophilia?

- A)** 0 **B)** 1/4 **C)** all of them **D)** 1/2 **E)** 3/4

Answer: D

Given: mother $X^H X^h$ (carrier), father $X^h Y$ (affected).

Relationship: every daughter receives the father's only X chromosome (X^h); she additionally receives X^H or X^h from the mother with probability one half each.

Substitute: daughter genotypes are $X^H X^h$ (carrier) with probability 1/2 and $X^h X^h$ (affected) with probability 1/2.

Result: 1/2 of the daughters are expected to have haemophilia.

Classic slip: 1/4 is the fraction among all children, because it folds in the separate 1/2 chance of being a daughter at all; the question asks only about daughters.

13. In double-stranded DNA, guanine pairs with cytosine through:

- A)** three hydrogen bonds
- B)** one hydrogen bond
- C)** four hydrogen bonds
- D)** a covalent bond
- E)** two hydrogen bonds

Answer: A

The decisive fact: a guanine-cytosine pair is joined by three hydrogen bonds, while an adenine-thymine pair has only two. This is why GC-rich DNA needs more heat to separate. The bonds are hydrogen bonds, not covalent links between the strands.

14. Retroviruses such as HIV copy their RNA genome into DNA inside the host cell using:

- A)** DNA polymerase of the host
- B)** restriction enzymes
- C)** ribozymes
- D)** RNA polymerase
- E)** reverse transcriptase

Answer: E

Retroviruses carry reverse transcriptase, an RNA-dependent DNA polymerase that copies the viral RNA genome into DNA, which then integrates into the host chromosome. The host's own DNA polymerase copies DNA templates, not RNA, and RNA polymerase makes RNA from DNA, the wrong direction for this step.

15. During translation, the mRNA codon 5'-AUG-3' is read by the tRNA anticodon. Written in the 3' to 5' direction, the anticodon is:

- A)** 3'-UAC-5' **B)** 3'-AUG-5' **C)** 3'-TAC-5' **D)** 5'-UAC-3' **E)** 5'-ATG-3'

Answer: A

Codon and anticodon pair antiparallel. Reading the codon 5'-AUG-3', the pairing runs A with U, U with A, G with C, so the anticodon written in the 3' to 5' direction is 3'-UAC-5'. Written in the 5' to 3' direction the same anticodon would read CAU, and any sequence containing T cannot be right because tRNA is RNA.

16. Which protein exhibits quaternary structure?

- A)** ribonuclease, a single 124-residue chain
- B)** pepsin, a single-chain enzyme
- C)** haemoglobin, with two α and two β chains
- D)** myoglobin, a single 153-residue chain
- E)** an α -helix

Answer: C

Quaternary structure means an assembly of more than one polypeptide chain. Haemoglobin, with its four subunits (two alpha and two beta chains), is the textbook example. Myoglobin, pepsin and ribonuclease are each a single chain and so show at most tertiary structure, while an alpha-helix is a motif of secondary structure, not a whole protein organisation.

17. When two amino acids join to form a dipeptide, the reaction is:

- A)** hydrolysis, consuming water
- B)** a condensation reaction releasing one molecule of water and forming a peptide bond
- C)** a substitution of the R groups
- D)** a neutralisation releasing salt
- E)** an oxidation releasing hydrogen

Answer: B

The amino group of one amino acid attacks the carboxyl group of the other; one molecule of water is released and a peptide (amide) bond forms. This is a condensation reaction. Hydrolysis is the reverse process, breaking the bond by adding water, so the option describing hydrolysis with water consumed gets the direction of the reaction backwards.

18. Fermentation yields only about 2 ATP per glucose, versus roughly 30 by full aerobic oxidation, because:

- A)** NAD^+ is never regenerated in fermentation
- B)** the electron transport chain runs backwards during fermentation
- C)** glycolysis stops when fermentation begins
- D)** fermentation produces no ATP at all
- E)** the glucose is only partially oxidised: most of its energy remains trapped in lactate or ethanol

Answer: E

Fermentation is glycolysis plus the regeneration of NAD^+ , and it stops at reduced end-products such as lactate or ethanol. The glucose carbon is never fully oxidised to CO_2 and water, and the Krebs cycle and electron transport chain never run, so almost all of the glucose's energy remains locked in the end-products. The 2 ATP that are made all come from glycolysis, so fermentation certainly does produce ATP.

19. Substrate-level phosphorylation (direct transfer of a phosphate from a substrate to ADP) occurs in:

- A)** pyruvate oxidation only
- B)** the electron transport chain only
- C)** both glycolysis and the Krebs cycle
- D)** glycolysis only
- E)** the Krebs cycle only

Answer: C

Substrate-level phosphorylation is direct transfer of a phosphate group from a substrate to ADP. It occurs at two steps of glycolysis (the phosphoglycerate kinase and pyruvate kinase steps) and at one step of the Krebs cycle (succinyl-CoA synthetase). The electron transport chain instead uses oxidative phosphorylation through a proton gradient, which is why the option naming it is wrong.

20. Leaves look green because chlorophyll:

- A)** is a green pigment that absorbs all wavelengths equally
- B)** destroys green light
- C)** absorbs only green light
- D)** absorbs red and blue light strongly and reflects/transmits green light
- E)** emits green fluorescence under sunlight

Answer: D

Chlorophyll's absorption peaks strongly in the blue and red regions of the spectrum; green light is largely reflected and transmitted, so that is the light reaching our eyes and leaves look green. The claim that it absorbs only green light inverts the truth (leaves would then appear the complementary colour), and absorbing all wavelengths equally would make leaves look black.

21. The sodium-potassium pump of the cell membrane moves:

- A)** 3 Na^+ out for every 2 K^+ in, at the cost of one ATP
- B)** 2 Na^+ out for every 3 K^+ in
- C)** Na^+ and K^+ both down their gradients
- D)** 3 K^+ out for every 2 Na^+ in
- E)** equal numbers of Na^+ and K^+ in opposite directions

Answer: A

The sodium-potassium pump (Na^+/K^+ -ATPase) hydrolyses one ATP to export 3 Na^+ and import 2 K^+ , each ion moved against its gradient. This electrogenic imbalance (net loss of one positive charge per cycle) contributes to the resting potential and keeps the ionic gradients that nerve and muscle firing consume. The ratio 3 out : 2 in at a cost of one ATP is the only version matching the textbook pump.

22. Which checkpoint ensures that DNA replication is complete and undamaged before mitosis begins?
- A) the M checkpoint (spindle)
 - B) the G2 checkpoint
 - C) the S checkpoint
 - D) cytokinesis
 - E) the G1 checkpoint

Answer: B

The G2 checkpoint sits after S phase and before mitosis, and it verifies that DNA replication is complete and that any damage is repaired before the cell commits to division. The G1 checkpoint decides whether to begin a new cycle (cell size, nutrients, DNA integrity at the start), and the spindle checkpoint acts inside mitosis at metaphase, too late to check replication.

23. Meiosis II resembles mitosis because in both processes:
- A) the cell becomes haploid for the first time
 - B) bivalents align on the equator
 - C) sister chromatids separate and move to opposite poles
 - D) homologous chromosomes pair and cross over
 - E) the chromosome number is halved

Answer: C

In both meiosis II and mitosis, sister chromatids separate and move to opposite poles, so the mechanics of segregation are the same. Pairing of homologues, crossing over, halving of the chromosome number and alignment of bivalents on the equator are all features of meiosis I, and the cell first becomes haploid at the end of meiosis I, already before meiosis II begins.

24. The inside of a resting neurone is negative relative to the outside mainly because:
- A) the membrane is impermeable to all ions at rest
 - B) chloride ions enter the cell continuously
 - C) large anions are actively transported out
 - D) potassium ions diffuse out through leak channels down their gradient, leaving unbalanced negative charge inside
 - E) sodium ions are pumped into the cell

Answer: D

At rest the membrane is most permeable to potassium through open K⁺ leak channels. Potassium diffuses out down its concentration gradient, leaving behind unbalanced intracellular negative charge (large fixed anions that cannot follow), and this builds the resting potential of about minus 70 mV. The pump maintains the gradients over time, but its 3-out : 2-in imbalance is only a minor direct contributor to the voltage.

25. Which sequence correctly traces a simple reflex arc?
- A) receptor → CNS → effector → motor neurone → sensory neurone
 - B) effector → motor neurone → CNS → sensory neurone → receptor
 - C) receptor → motor neurone → CNS → sensory neurone → effector
 - D) CNS → receptor → sensory neurone → motor neurone → effector
 - E) receptor → sensory neurone → CNS → motor neurone → effector

Answer: E

A reflex arc runs: receptor detects the stimulus, sensory (afferent) neurone carries the signal to the CNS, integration occurs in the centre, motor (efferent) neurone carries the command outward, and the effector (muscle or gland) responds. Every sequence that puts the motor neurone before the CNS, or the CNS before the receptor, reverses the direction of information flow.

26. Which hormone raises blood glucose as part of the acute stress response?

- A)** ADH (vasopressin)
- B)** insulin
- C)** oxytocin
- D)** aldosterone
- E)** adrenaline (epinephrine)

Answer: E

The decisive fact: adrenaline, released from the adrenal medulla as part of the acute fight-or-flight response, stimulates glycogenolysis in the liver and raises blood glucose to fuel rapid action. Insulin lowers blood glucose (the opposite direction), ADH and aldosterone regulate water and sodium balance rather than glucose, and oxytocin concerns labour and milk ejection.

27. Vaccination with a live attenuated (weakened) pathogen produces which type of immunity?

- A)** active artificial immunity, with the person's own plasma cells and memory cells
- B)** natural passive immunity via the placenta
- C)** natural active immunity after catching the disease
- D)** passive artificial immunity from the injected antigens themselves
- E)** no immunity, since the pathogen is weakened

Answer: A

A live attenuated vaccine delivers an artificially introduced but weakened microbe that multiplies slightly and provokes a genuine primary immune response, generating the person's own plasma cells and memory cells: active, artificially acquired immunity. Passive immunity would mean receiving ready-made antibodies, which is not what a vaccine injects, and the fact that the pathogen is weakened does not stop it from being immunogenic, which is the whole point of attenuation.

28. The natural pacemaker region that initiates each heartbeat is the:

- A)** bundle of His
- B)** vagus nerve
- C)** sinoatrial (SA) node
- D)** atrioventricular (AV) node
- E)** Purkinje fibres

Answer: C

The sinoatrial node, a small region in the wall of the right atrium, depolarises fastest of all cardiac cells and initiates each heartbeat, making it the natural pacemaker. The atrioventricular node is only the normal relay to the ventricles, and the bundle of His and Purkinje fibres conduct the impulse onward; the vagus nerve modulates the rate but does not set the rhythm.

29. Gas exchange in the pulmonary alveoli is highly efficient mainly because:

- A)** the alveoli are muscular and pump air
- B)** each alveolus contains cartilage rings
- C)** the alveoli store oxygen like a tank
- D)** their enormous total surface area is combined with an extremely thin (squamous) wall
- E)** alveolar air is under high pressure

Answer: D

Diffusion efficiency scales with surface area divided by wall thickness. Hundreds of millions of alveoli provide roughly 70 square metres of exchange surface behind walls of extremely thin squamous cells (about half a micrometre), which is the structural basis of their efficiency. Alveoli have no muscle to pump air, no cartilage rings (those are in the larger airways), and they store no oxygen.

30. ADH (vasopressin) increases water reabsorption in the kidney by:

- A)** constricting the efferent arteriole
- B)** inserting aquaporin channels into the collecting duct wall
- C)** opening sodium channels in the loop of Henle
- D)** raising glomerular filtration pressure
- E)** dissolving urea in the filtrate

Answer: B

ADH binds V2 receptors on collecting duct cells and triggers insertion of aquaporin-2 channels into their membrane, so water leaves the filtrate down the medullary osmotic gradient and urine becomes concentrated. It does not act by changing filtration pressure, opening sodium channels, or constricting the efferent arteriole; it is a water-permeability hormone.

31. PCR (the polymerase chain reaction) requires a heat-stable DNA polymerase because:

- A)** the enzyme must work at refrigerator temperature
- B)** DNA cannot be copied at low temperatures
- C)** the primers are destroyed by heat and must be resynthesised
- D)** heat is needed to insert the product into bacteria
- E)** each cycle includes a high-temperature step (about 95 °C) that separates the DNA strands

Answer: E

Each PCR cycle starts with a denaturation step near 95 degrees Celsius that separates the DNA strands, and this step repeats dozens of times. An ordinary polymerase would be inactivated after the first few cycles, so a heat-stable polymerase such as Taq (from hot-spring bacteria) is needed to survive the whole programme. It is the high-temperature strand-separation step, not low temperature, that demands the special enzyme.

32. An antibiotic blocks the bacterial 70S ribosome but not the eukaryotic 80S ribosome. This selectivity means the drug can:

- A)** work only against viruses
- B)** act on the patient's mitochondria only
- C)** harm bacteria without directly stopping protein synthesis in the patient's own cells
- D)** be given at any dose without side effects
- E)** kill all cells equally

Answer: C

This is selective toxicity: because bacterial ribosomes are 70S while the patient's own cytoplasmic ribosomes are 80S, the drug can shut down bacterial protein synthesis without directly stopping protein synthesis in the patient's cells. It is not side-effect free at any dose, since human mitochondria carry their own 70S-like ribosomes and can be secondarily affected, and the drug targets bacteria, not viruses or mitochondria alone.

Section C - Chemistry (Q33-47)

33. What is the oxidation number of carbon in methanol, CH_3OH ?

- A) -4 B) -2 C) 0 D) $+2$ E) $+4$

Answer: B

Given: methanol CH_3OH , a neutral molecule; standard values $\text{H} = +1$, $\text{O} = -2$.

Relationship: in a neutral molecule the oxidation numbers sum to 0.

Substitute: $\text{C} + 4 \times (+1) + (-2) = 0$

Arithmetic: $\text{C} + 4 - 2 = 0$, so $\text{C} + 2 = 0$

Result: carbon is at -2 .

Classic slip: $+4$ is the oxidation number of the carbon in CO_2 , reached by counting oxygens instead of hydrogens; $+2$ belongs to an aldehyde carbon, one oxidation step above methanol.

34. In the reaction $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaOCl} + \text{H}_2\text{O}$, the chlorine:

- A) is simultaneously oxidised and reduced (disproportionation): $0 \rightarrow -1$ in NaCl and $0 \rightarrow +1$ in NaOCl
B) is reduced only
C) is oxidised only
D) keeps oxidation number 0 in both products
E) is unchanged, only dissolved

Answer: A

Given: $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaOCl} + \text{H}_2\text{O}$; chlorine starts as the element, oxidation number 0; Na is $+1$ and O is -2 throughout the products.

In NaCl the chlorine sits at -1 (plain chloride).

In NaOCl : $+1 + \text{Cl} + (-2) = 0$, so $\text{Cl} = +1$

One chlorine atom has gone $0 \rightarrow -1$ (gained an electron, reduced); the other has gone $0 \rightarrow +1$ (lost an electron, oxidised).

Result: the same element is oxidised and reduced in the same reaction, a disproportionation.

The tempting reduced-only or oxidised-only readings each cover just one of the two products; unchanged fails because the oxidation numbers demonstrably leave 0.

35. Which molecule has zero net dipole moment despite having polar bonds?

- A) NH_3 B) SO_2 C) H_2O D) BF_3 E) CHCl_3

Answer: D

BF_3 is trigonal planar: three identical polar B-F bonds point symmetrically at 120 degrees, so their dipoles cancel vectorially and the molecule has zero net dipole despite the polar bonds. In NH_3 (trigonal pyramidal), H_2O and SO_2 (both bent) and CHCl_3 (an asymmetric tetrahedron) the bond dipoles do not cancel, leaving a permanent net dipole.

36. Which molecule has a trigonal pyramidal shape?

- A) H_2O B) NH_3 C) BF_3 D) CO_2 E) CH_4

Answer: B

NH_3 has three bonding pairs plus one lone pair on nitrogen. The tetrahedral electron arrangement leaves the molecular shape trigonal pyramidal with a bond angle of about 107 degrees. CH_4 (no lone pairs) is tetrahedral, H_2O (two lone pairs) is bent, BF_3 is trigonal planar and CO_2 is linear.

37. What is the pH of a 0.005 M solution of $\text{Ba}(\text{OH})_2$ at 25 °C? ($\text{Ba}(\text{OH})_2$ releases 2 OH^- per formula unit; $K_w = 1.0 \times 10^{-14}$)

- A)** 1.7 **B)** 2 **C)** 11 **D)** 12 **E)** 13

Answer: D

Given: $c = 0.005 \text{ M Ba}(\text{OH})_2$; each formula unit releases 2 OH^- ; $K_w = 1.0 \times 10^{-14}$, so $\text{pH} + \text{pOH} = 14$ at 25 °C.

Relationship: $[\text{OH}^-] = 2 \times c$; $\text{pOH} = -\log_{10}[\text{OH}^-]$; $\text{pH} = 14 - \text{pOH}$

Substitute: $[\text{OH}^-] = 2 \times 0.005 = 0.010 \text{ mol/L}$

$\text{pOH} = -\log(0.010) = 2$

$\text{pH} = 14 - 2 = 12$

Result: pH 12.

Classic slip: ignoring the factor 2 gives $[\text{OH}^-] = 0.005 \text{ M}$, $\text{pOH} \approx 2.3$ and $\text{pH} \approx 11.7$, the tempting value near 11; and the intermediate 2 is the pOH, not the pH.

38. Solution A has pH 4 and solution B has pH 6. The $[\text{H}^+]$ of A compared with B is:

- A)** 1.5 times greater
B) 2 times greater
C) 10 times greater
D) 100 times greater
E) 1000 times greater

Answer: D

Given: solution A at pH 4, solution B at pH 6.

Relationship: $\text{pH} = -\log_{10}[\text{H}^+]$, so each 1-unit step in pH is a factor of 10 in hydrogen ion concentration.

Substitute: $[\text{H}^+]$ of A = 10^{-4} mol/L ; $[\text{H}^+]$ of B = 10^{-6} mol/L

Ratio: $10^{-4} \div 10^{-6} = 10^2 = 100$

Result: A's hydrogen ion concentration is 100 times greater than B's.

Classic slip: 2 times greater reads the 2-unit pH gap as a linear factor, but the scale is logarithmic.

39. How many atoms (total) are present in 0.25 mol of CO_2 ? ($N_a = 6.02 \times 10^{23} \text{ mol}^{-1}$)

- A)** 0.75×10^{23} **B)** 1.5×10^{23} **C)** 3.0×10^{23} **D)** 4.5×10^{23} **E)** 6.0×10^{23}

Answer: D

Given: $n = 0.25 \text{ mol of CO}_2$; $N_a = 6.02 \times 10^{23} \text{ mol}^{-1}$; each CO_2 molecule contains 3 atoms (1 carbon + 2 oxygen).

Relationship: molecules = $n \times N_a$, then total atoms = molecules $\times 3$.

Substitute: $0.25 \times 6.02 \times 10^{23} = 1.5 \times 10^{23} \text{ molecules}$

Multiply by 3: $3 \times 1.5 \times 10^{23} = 4.5 \times 10^{23} \text{ atoms}$

Result: 4.5×10^{23} atoms in total.

Classic slip: stopping at 1.5×10^{23} reports molecules, not atoms; counting only one oxygen ($\times 2$ per molecule) gives 3.0×10^{23} .

40. Heating 10.0 g of pure CaCO_3 ($M = 100 \text{ g/mol}$) would release 2.24 L of CO_2 at STP. An experiment on an impure sample of the same mass releases 1.68 L at STP. The percentage of CaCO_3 in the sample is:

- A)** 25% **B)** 60% **C)** 67% **D)** 75% **E)** 133%

Answer: D

Given: 10.0 g of pure CaCO_3 ($M = 100 \text{ g/mol}$) releases 2.24 L of CO_2 at STP; the impure 10.0 g sample releases 1.68 L at STP.

Relationship: CO_2 volume is directly proportional to the amount of CaCO_3 present, so the purity fraction equals the volume ratio.

Substitute: $\text{purity} = 1.68 \div 2.24 = 0.75$

Convert: $0.75 \times 100 = 75 \text{ percent}$

Result: the sample is 75 percent CaCO_3 .

Classic slip: 25 percent divides the missing volume $0.56 \div 2.24$ and so measures the impurity, not the carbonate; 133 percent inverts the ratio and is impossible for a percentage.

41. A gas at 27°C and 1.0 atm is sealed in a rigid vessel and heated to 127°C . The new pressure is closest to:

- A)** 0.75 atm **B)** 1.0 atm **C)** 1.11 atm **D)** 1.33 atm **E)** 4.2 atm

Answer: D

Given: $P_1 = 1.0 \text{ atm}$; $T_1 = 27^\circ\text{C}$; $T_2 = 127^\circ\text{C}$; rigid vessel, so volume and amount are constant.

Relationship: Gay-Lussac's law, $P_2 = P_1 \times T_2 \div T_1$, valid only in absolute temperature.

Convert: $T_1 = 27 + 273 = 300 \text{ K}$; $T_2 = 127 + 273 = 400 \text{ K}$

Substitute: $P_2 = 1.0 \times 400 \div 300$

Arithmetic: $400 \div 300 = 1.33$

Result: the new pressure is about 1.33 atm.

Classic slip: inverting the ratio ($300 \div 400$) predicts a pressure fall from heating, which is unphysical; dividing 127 by 27 in plain Celsius is meaningless because the Celsius zero is arbitrary.

42. A chlorine atom is smaller than a sodium atom mainly because:

- A)** chlorine has more protons pulling on electrons in the same shell
B) chlorine gains electrons, sodium loses them
C) sodium is a metal and chlorine is a gas
D) chlorine's electrons repel each other more strongly
E) chlorine has more electron shells

Answer: A

Sodium and chlorine belong to the same period, so both have their outer electrons in the same third shell. Chlorine's nucleus carries 17 protons against sodium's 11, and this greater nuclear charge pulls the same-shell electrons closer, making the atom smaller. The comparison concerns neutral atoms, so ion formation is irrelevant, and the shell count is identical, not higher, for chlorine.

43. Methanol reacts with propanoic acid under reflux with concentrated H_2SO_4 . The organic product is:
- A) propanal
 - B) methanoic acid
 - C) propanoic anhydride
 - D) propan-1-ol
 - E) methyl propanoate

Answer: E

An alcohol and a carboxylic acid under reflux with concentrated sulfuric acid undergo esterification. The alcohol supplies the alkyl part (methanol gives methyl) and the acid supplies the carboxylate part (propanoic acid gives propanoate), so the product is methyl propanoate, $\text{CH}_3\text{CH}_2\text{COOCH}_3$, with water as by-product. The reverse name (propanyl-type esters) or any aldehyde or anhydride misreads which fragment comes from which reagent.

44. Hydrogen bromide adds to propene ($\text{CH}_3\text{-CH=CH}_2$). The major product is:
- A) propane
 - B) 1,2-dibromopropane
 - C) 2-bromopropane
 - D) 2-bromopropene
 - E) 1-bromopropane

Answer: C

Hydrogen bromide adds to propene by Markovnikov's rule: H attaches to the double-bond carbon that already has more hydrogens (the terminal CH_2), because that route passes through the more stable secondary carbocation, and Br then bonds to the middle carbon. The product is 2-bromopropane, $\text{CH}_3\text{-CHBr-CH}_3$. The bromine on the end carbon (1-bromopropane) would be the anti-Markovnikov product, obtained only in the presence of peroxides, and the dibromide would need Br_2 rather than HBr .

45. For the heterogeneous equilibrium $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$, the equilibrium constant is written as:
- A) $K_c = [\text{CaO}][\text{CO}_2]$
 - B) $K_c = 1$, since solids are constant
 - C) $K_c = [\text{CO}_2]$
 - D) $K_c = 1/[\text{CO}_2]$
 - E) $K_c = [\text{CO}_2]/([\text{CaO}][\text{CaCO}_3])$

Answer: C

In a heterogeneous equilibrium the pure solids appear in fixed amounts, so their terms are constants that are folded into the equilibrium constant and omitted from the expression.

Only the gaseous product remains: $K_c = [\text{CO}_2]$

Including the solids in a quotient, or inverting the expression, misapplies the rule that pure phases are excluded.

46. In a working galvanic (voltaic) cell:
- A) anions in the salt bridge migrate toward the cathode compartment
 - B) electrons flow from cathode to anode through the external wire
 - C) oxidation occurs at the cathode
 - D) electrons flow through the salt bridge
 - E) oxidation occurs at the anode and the electrons released flow through the external wire from anode to cathode

Answer: E

In any working galvanic cell, oxidation occurs at the anode, and the electrons released travel through the external wire from anode to cathode, where reduction takes place. The salt bridge carries ions, never electrons, and its anions migrate toward the anode compartment to neutralise the positive charge building up there (cations drift toward the cathode side). Any statement that puts oxidation at the cathode, or electrons in the bridge, contradicts these roles.

47. Four 0.10 M solutions are prepared: glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), NaCl, CaCl_2 , and sucrose. Which has the lowest freezing point?

- A) CaCl_2 (three ions per formula unit)
- B) sucrose (no ions)
- C) NaCl (two ions)
- D) all four freeze at the same temperature
- E) glucose (no ions)

Answer: A

Given: four solutions, each 0.10 M: glucose $\text{C}_6\text{H}_{12}\text{O}_6$, NaCl, CaCl_2 , sucrose.

Relationship: freezing-point depression is colligative, set by the number of dissolved particles per litre, not their identity.

Count particles: glucose and sucrose do not ionise, 0.10 mol of particles per litre each.

NaCl gives 2 ions per formula unit: $2 \times 0.10 = 0.20$ mol of particles per litre.

CaCl_2 gives 3 ions per formula unit: $3 \times 0.10 = 0.30$ mol of particles per litre.

Result: the greatest particle count gives the greatest depression, so the CaCl_2 solution freezes lowest.

Classic slip: all four freezing together assumes equal molarity means equal effect, but only particle counts decide; NaCl's 0.20 loses to CaCl_2 's 0.30.

Section D - Mathematics (Q48-54)

48. Solve the inequality $8 - 5x < -12$.

- A) $x < -4$
- B) $x > 4$
- C) $x < 4$
- D) $x \geq 4$
- E) $x > -4$

Answer: B

Given: $8 - 5x < -12$.

Isolate the x term: subtract 8 from both sides, $-5x < -12 - 8$, so $-5x < -20$.

Divide both sides by -5 : division by a negative number flips the direction, so $x > 4$.

Check: at $x = 5$, $8 - 5 \times 5 = 8 - 25 = -17$, and $-17 < -12$ holds

Result: $x > 4$.

Classic slip: dividing by -5 without flipping gives $x < 4$, the exact reverse set; $x = 4$ itself fails because $8 - 20 = -12$ is not strictly less than -12 , so the loose bound of 4 or more is wrong too.

49. Solve $-3 < 2x + 1 \leq 7$.

- A) $-4 < x \leq 6$ B) $-2 \leq x \leq 3$ C) $-2 < x < 3$ D) $-2 < x \leq 3$ E) $-1 < x \leq 4$

Answer: D

Given: $-3 < 2x + 1 \leq 7$.

Subtract 1 from all three parts: $-4 < 2x \leq 6$.

Divide all three parts by 2, a positive number, so both directions and strictness survive: $-2 < x \leq 3$.

Result: $-2 < x \leq 3$.

Classic slip: making the left bound inclusive ($-2 \leq x$) forgets that the original -3 was strict; testing $x = -2$ gives -3 on the left, which fails.

50. If $2^x = 40$, then x satisfies:

- A) $x = 6$ B) $4 < x < 5$ C) $6 < x < 7$ D) $x = 5$ E) $5 < x < 6$

Answer: E

Given: $2^x = 40$

Relationship: 2^x rises continuously with x , so bracket 40 between neighbouring whole-number powers.

Compute: $2^5 = 32$ and $2^6 = 64$

Compare: $32 < 40 < 64$, so x sits strictly between 5 and 6.

Result: $5 < x < 6$ (exactly $\log_2 40 \approx 5.32$).

Classic slip: $x = 5$ or $x = 6$ would require 40 to equal 32 or 64, which it does not.

51. One card is drawn at random from a standard 52-card deck. What is the probability it is a heart or a king?

- A) $1/4$ B) $1/13$ C) $4/13$ D) $13/52$ E) $17/52$

Answer: C

Given: one card drawn from a standard 52-card deck; wanted: heart or king.

Relationship: inclusion-exclusion, favourable = hearts + kings – cards counted in both sets.

Count: 13 hearts; 4 kings; the king of hearts lies in both, so subtract 1.

Favourable = $13 + 4 - 1 = 16$

Probability = $16 \div 52 = 4/13$

Result: $4/13$.

Classic slip: $17/52$ adds $13 + 4$ and never removes the king of hearts, which is simultaneously a heart and a king.

52. A straight ramp 8 m long makes a 30° angle with the horizontal. How high is its upper end above the ground? ($\sin 30^\circ = 0.5$, $\cos 30^\circ = 0.87$)

- A)** 2 m **B)** 4 m **C)** 6.9 m **D)** 8 m **E)** 16 m

Answer: B

Given: ramp length (hypotenuse) 8 m; angle with the horizontal 30° ; $\sin 30^\circ = 0.5$, $\cos 30^\circ = 0.87$.

Relationship: in a right triangle the side opposite an angle equals hypotenuse $\times \sin(\text{angle})$; the height is opposite the 30° angle.

Substitute: height = $8 \times \sin 30^\circ = 8 \times 0.5$

Arithmetic: $8 \times 0.5 = 4$

Result: the upper end of the ramp is 4 m above the ground.

Classic slip: 6.9 m comes from $8 \times \cos 30^\circ = 8 \times 0.87$, which is the ramp's horizontal reach along the ground, not the vertical rise.

53. If $f(x) = 2x - 1$, then $f(f(x))$ equals:

- A)** $2x - 3$ **B)** $3x - 4$ **C)** $4x - 3$ **D)** $4x - 1$ **E)** $4x + 1$

Answer: C

Given: $f(x) = 2x - 1$; wanted $f(f(x))$

Relationship: composition feeds the inner output back into f as the input.

Substitute: $f(f(x)) = 2 \times (2x - 1) - 1$

Expand: $4x - 2 - 1$.

Collect: $4x - 3$.

Result: $f(f(x)) = 4x - 3$

Classic slip: $2x - 3$ doubles only once; both the doubling and the subtraction happen at each of the two passes through f .

54. The circle $(x - 1)^2 + (y + 2)^2 = 16$ has centre and radius:

- A)** (1, -2), radius 4
B) (1, -2), radius 16
C) (-1, 2), radius 4
D) (-1, 2), radius 16
E) (1, 2), radius 4

Answer: A

Given: $(x - 1)^2 + (y + 2)^2 = 16$

Relationship: standard circle form $(x - h)^2 + (y - k)^2 = r^2$, centre (h, k) , radius r .

Match: $x - 1 = x - h$ gives $h = 1$; $y + 2 = y - (-2)$ gives $k = -2$

Radius: $r^2 = 16$, so $r = \sqrt{16} = 4$

Result: centre (1, -2), radius 4.

Classic slip: reading the plus sign in $y + 2$ as $k = +2$ puts the centre in the wrong quadrant, and quoting 16 as the radius confuses r^2 with r .

Section E - Physics (Q55-60)

55. A 24 V battery drives a 2 Ω resistor in series with a parallel pair of 12 Ω and 6 Ω . What current flows from the battery?

- A) 2 A B) 3 A C) 4 A D) 6 A E) 12 A

Answer: C

Given: $V = 24$ V battery; 2 Ω resistor in series with a parallel pair of 12 Ω and 6 Ω .

Relationship: parallel pair via $1/R = 1/R_1 + 1/R_2$; series resistances add; Ohm's law $I = V \div R$.

Substitute: $1/R = 1/12 + 1/6$; with $1/6 = 2/12$ this gives $1/R = 3/12$

Invert: $R = 12/3 = 4$ Ω

Add the series resistor: $2 + 4 = 6$ Ω total

Substitute into Ohm's law: $I = 24 \div 6 = 4$ A

Result: 4 A flows from the battery.

Classic slip: 2 A comes from $24 \div 12$, which ignores the parallel section; 6 A comes from $24 \div 4$, which forgets the series 2 Ω .

56. A 60 W lamp is designed for a 120 V supply. The resistance of its filament is:

- A) 0.5 Ω B) 2 Ω C) 60 Ω D) 240 Ω E) 7200 Ω

Answer: D

Given: $P = 60$ W, $V = 120$ V

Relationship: $P = V^2 \div R$, rearranged as $R = V^2 \div P$

Substitute: $R = 120^2 \div 60$

Square: $120^2 = 14400$

Divide: $14400 \div 60 = 240$

Result: the filament resistance is 240 Ω .

Classic slip: 2 Ω comes from $V \div P = 120 \div 60$, which actually computes the current the lamp draws, not the resistance.

57. Water boiling at atmospheric pressure stays at exactly 100 °C even though the flame keeps supplying heat. This is because:

- A)** the supplied energy goes into the latent heat of vaporisation, breaking intermolecular attractions instead of raising temperature
- B)** boiling only occurs without heating
- C)** the thermometer is faulty above 90 °C
- D)** water has no specific heat at 100 °C
- E)** the molecules stop moving at 100 °C

Answer: A

Givens: water boiling at atmospheric pressure; flame supplies heat continuously; thermometer stays at exactly 100 °C.

Once boiling starts, added energy no longer increases the molecules' average kinetic energy, so the temperature cannot rise.

Every joule goes into the latent heat of vaporisation (about 2260 J per gram), pulling molecules apart against their intermolecular attractions.

A phase change in progress is isothermal: the reading stays pinned until the last liquid has evaporated.

Result: the temperature holds at 100 °C because the heat is consumed by the phase change itself.

The tempting claim that the molecules stop moving fails: they are in fact receiving the energy needed to break free as vapour.

58. Cars once had rigid metal bumpers; modern cars crumple in a crash. The physics reason a crumple zone reduces injury is that it:

- A)** eliminates the change of momentum
- B)** extends the collision time, reducing the average force for the same change of momentum
- C)** converts the passenger into the system's centre of mass
- D)** reduces the car's momentum to zero instantly
- E)** increases the impulse delivered

Answer: B

Givens: a crash fixes the change of momentum Δp the car must undergo; impulse relation $F \times \Delta t = \Delta p$.

The crumple zone deforms on impact, stretching the collision time Δt .

Rearrange: $F = \Delta p \div \Delta t$; with Δp fixed and Δt longer, the average force F shrinks.

Result: the same momentum change is delivered by a smaller average force on the occupants, and injury scales with force.

The tempting claims fail because the impulse is neither eliminated nor increased: Δp is set by the crash, only the time over which it is delivered changes.

59. On a distance-time graph, a horizontal segment means the object is:

- A)** moving at constant non-zero speed
- B)** decelerating to rest at the end of the segment
- C)** moving backwards
- D)** accelerating
- E)** stationary (zero speed for that interval)

Answer: E

Givens: a distance-time graph with a horizontal segment.

On a distance-time graph the gradient of the line is the speed, since $\text{speed} = \text{distance covered} \div \text{time taken}$.

A horizontal segment has zero gradient: zero rise in distance during the interval.

Zero distance covered means zero speed, so the object is stationary for that interval.

Result: a horizontal segment on a distance-time graph means the object is stationary.

The tempting reading of constant non-zero speed comes from importing the rule for velocity-time graphs, where a flat line does mean steady speed; read the axis label first.

60. A block of density 0.90 g/cm^3 floats in water (1.00 g/cm^3). What fraction of the block's volume is submerged?

- A)** 10% **B)** 25% **C)** 50% **D)** 90% **E)** 100%

Answer: D

Given: block density $\rho_{\text{block}} = 0.90 \text{ g/cm}^3$; water density $\rho_{\text{water}} = 1.00 \text{ g/cm}^3$; block volume V , submerged volume V_{sub} .

Relationship: floating equilibrium, weight = upthrust: $\rho_{\text{block}} \times V \times g = \rho_{\text{water}} \times V_{\text{sub}} \times g$.

Cancel g from both sides, then cancel V : $V_{\text{sub}} \div V = \rho_{\text{block}} \div \rho_{\text{water}} = 0.90 \div 1.00$.

Arithmetic: $0.90 \div 1.00 = 0.90$

Result: 0.90, so 90 percent of the block's volume is submerged.

Classic slip: 10 percent is the fraction above water, the complement of the submerged share; the submerged fraction tracks the density ratio itself.